

Surface-to>Returns

Visual Figure Package

Public-safe figures generated from non-raw reports and manifests.

SAMPLE 1996-2024 345 completed shards	SSVI GATES 100.0% 34,800 surfaces	CHARACTERISTICS 33 670,562 rows	SDF 0.116 227 OOS months	READINESS 15 pass 3 negative, 0 blocked
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Boundary: the latest return-prediction and portfolio evidence is weak/negative; external APIs are loaded only from safe runtime environment variables.

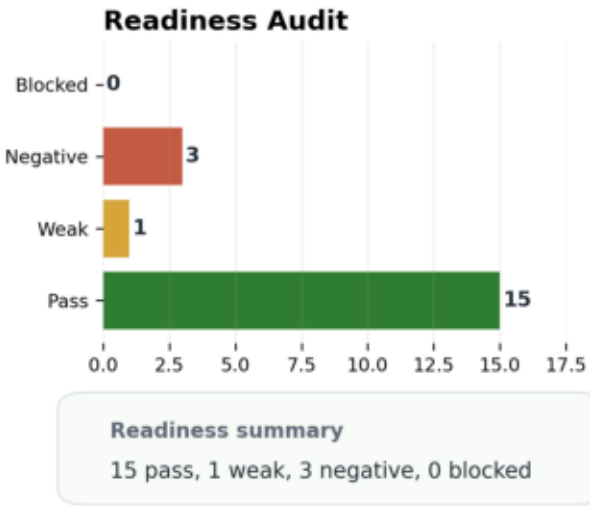
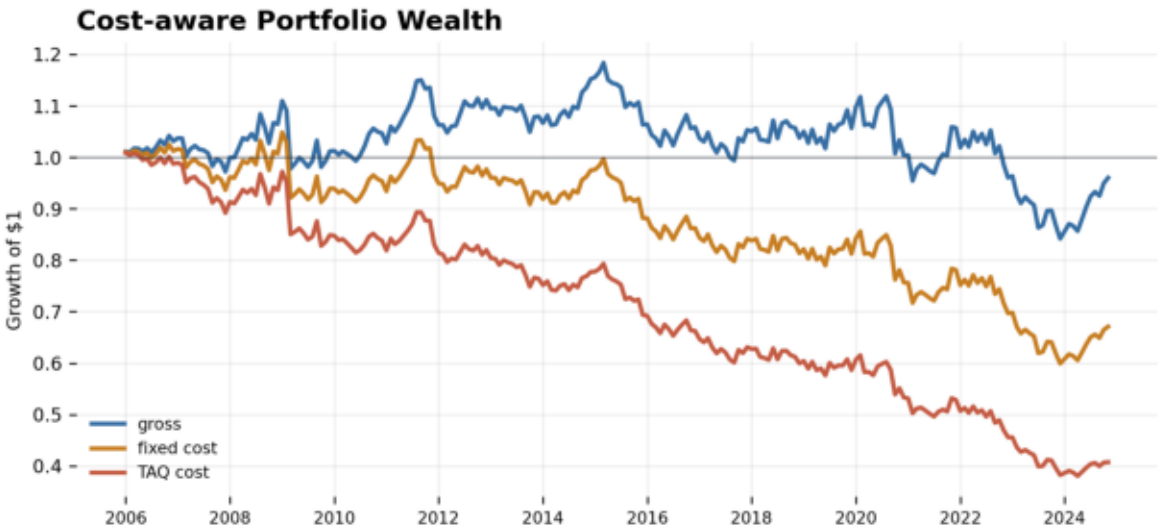
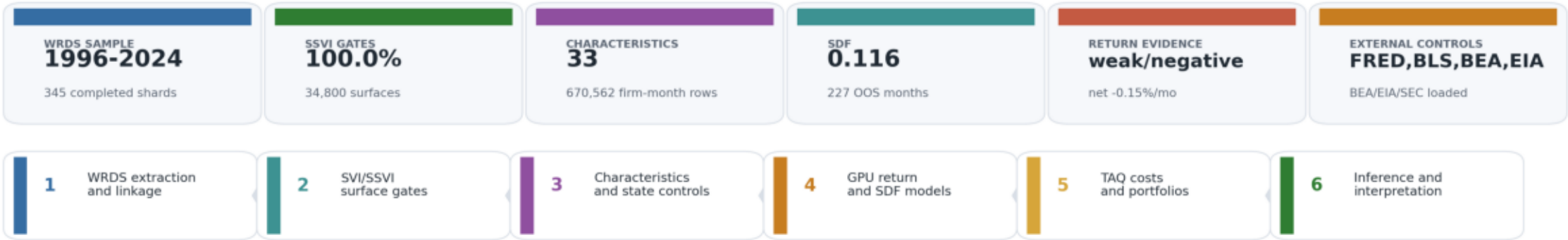
Visual Abstract

First-glance summary of the scaled implementation, no-arbitrage success, and honest return boundary.

Surface-to>Returns

Proposal-grade implementation evidence with no-arbitrage surface construction and disciplined return interpretation

Status: implementation-rich, alpha claim not supported



Honest Boundary

- Inference results are negative.
- Return prediction is weak.
- Cost-aware portfolio is negative.
- Reg SHO mechanism is weak.

What is proven: scaled construction, no-arbitrage diagnostics, full feature/state/cost/inference/model stack, and public-safe visuals.

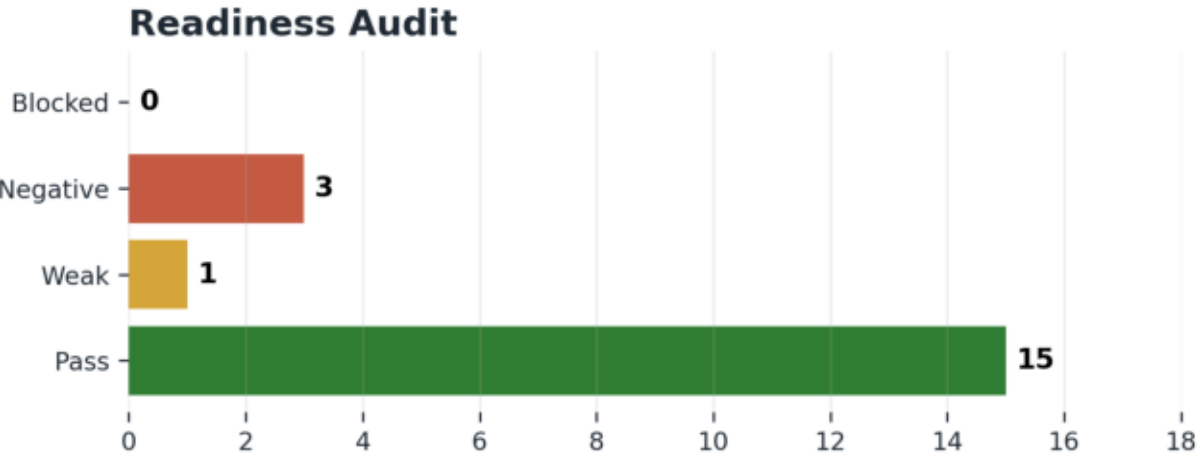
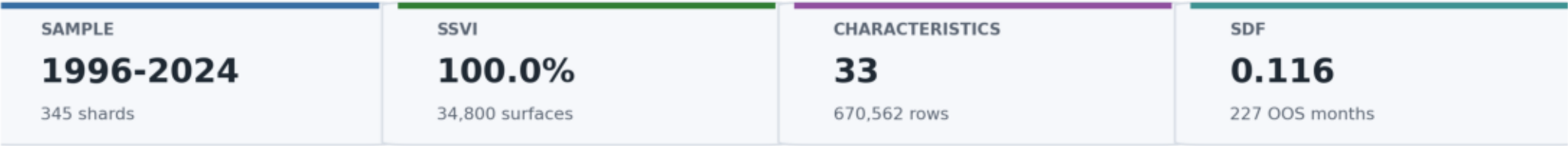
What is not proven: a positive return anomaly in the latest expanded-feature specification; external APIs are loaded only from safe runtime environment variables.

Figure 1. Data and Modeling Stack

WRDS extraction, linkage, surface construction, characteristics, costs, inference, and interpretation.

Figure 1. Proposal-Grade Data and Modeling Stack

WRDS extraction, no-arbitrage surface construction, characteristics, costs, inference, and interpretation are implemented at the usable 1996-2024 scale.



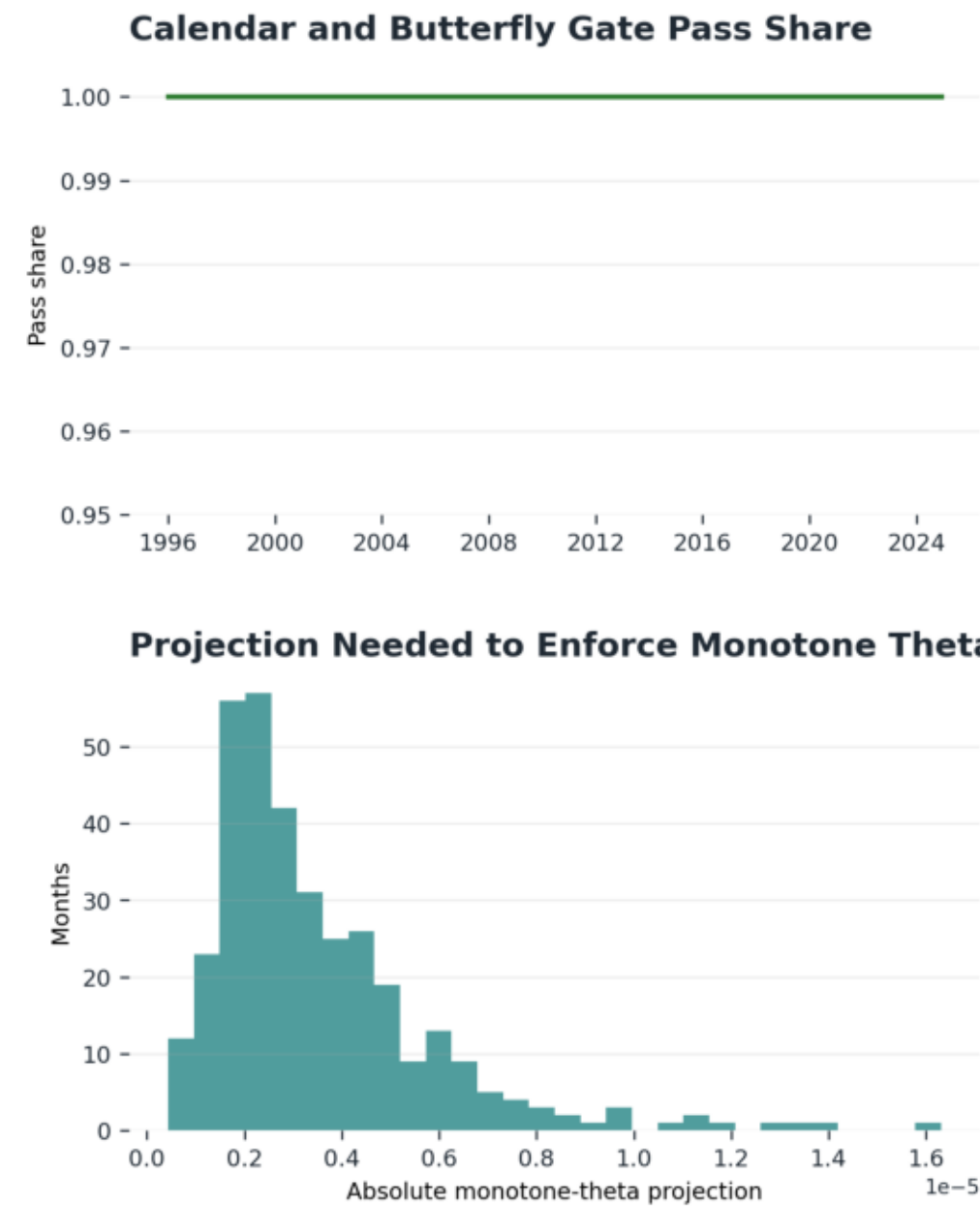
Evidence Boundary

- 2025 OptionMetrics shards were unavailable in this run.
- Return-prediction and portfolio evidence is weak to negative.
- FRED/BLS/BEA/EIA/SEC enrichment loaded from safe runtime env vars.

Figure 2. Surface Quality

SSVI pass share, fit error, monotone-theta projection, and no-arbitrage constraints.

Figure 2. SSVI Surface Quality and No-Arbitrage Diagnostics



Surface Evidence

- 34,800 of 34,800 date-security surfaces pass.
- Diagnostics enforce positivity, monotone theta, monotone theta-phi, calendar-grid monotonicity, and Gatheral-Jacquier bounds.
- Maximum theta projection in monthly report: 1.63e-05.

Figure 3. SDF and Interpretation

Three-branch conditional autoencoder SDF, integrated gradients, and TreeSHAP evidence.

Figure 3. Conditional Autoencoder SDF and Interpretation

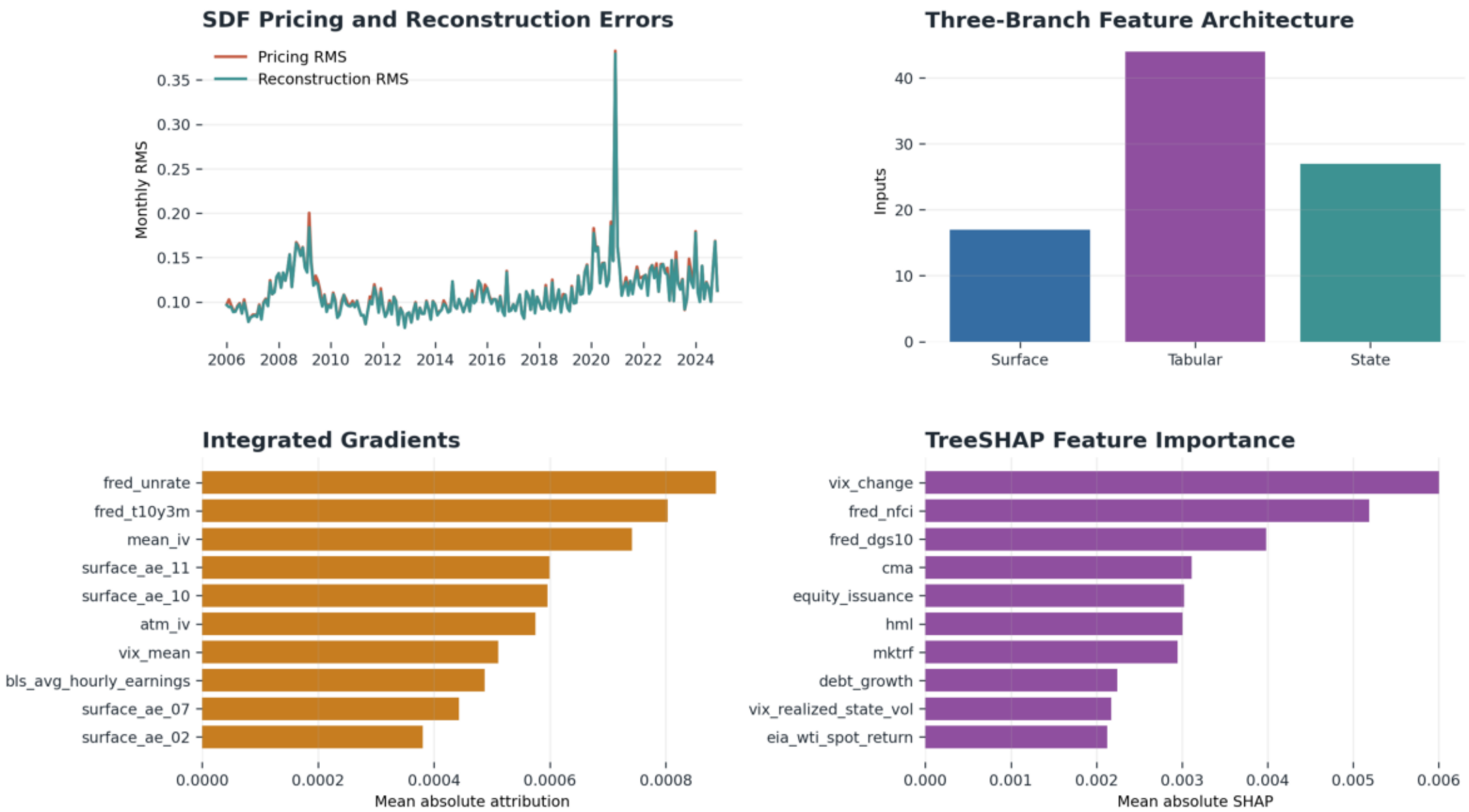


Figure 4. Return Evidence

Buffered long-short wealth, Sharpe ratios, factor alphas, and trading frictions.

Figure 4. Cost-Aware Return and Asset-Pricing Evidence

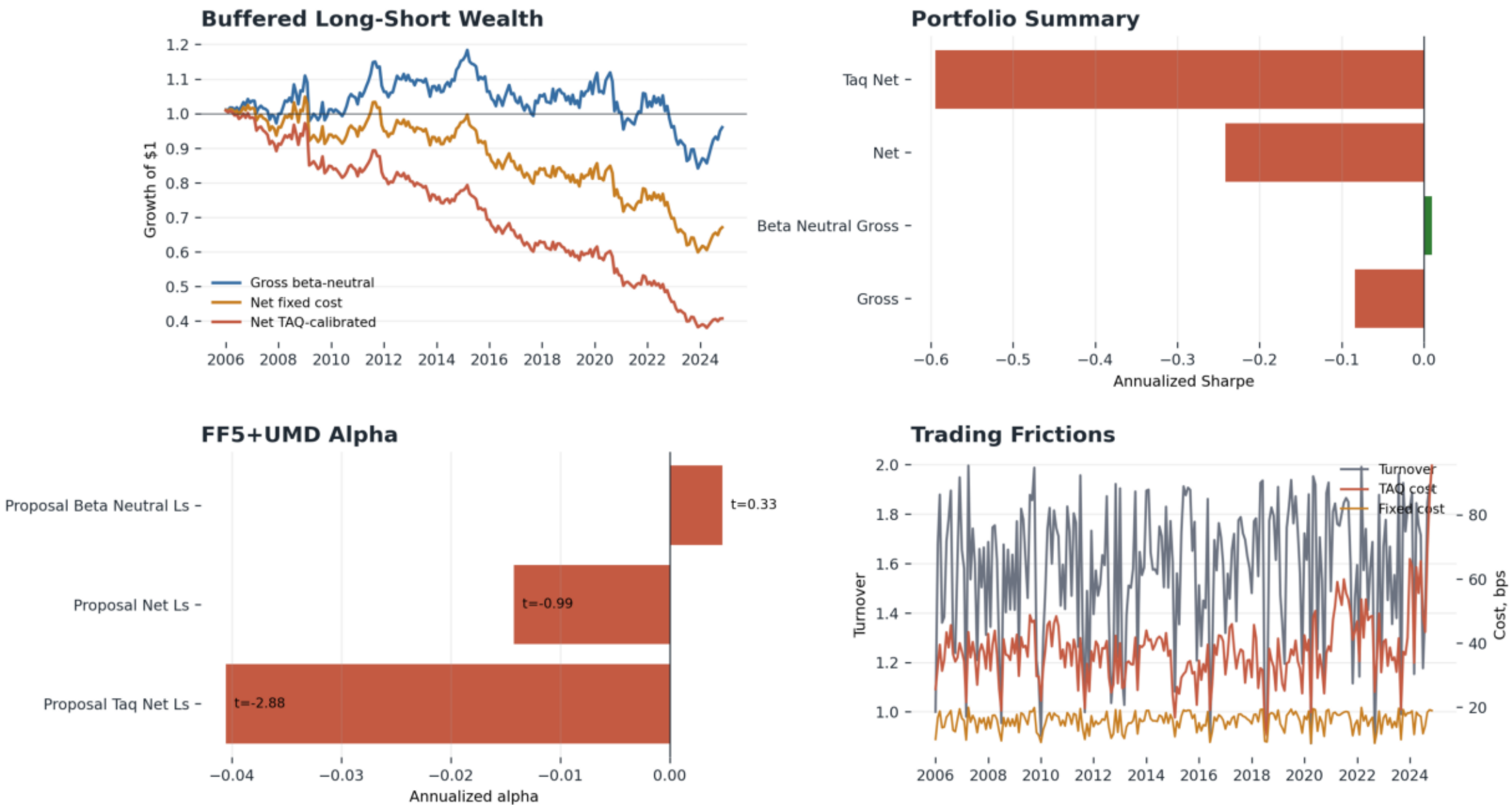
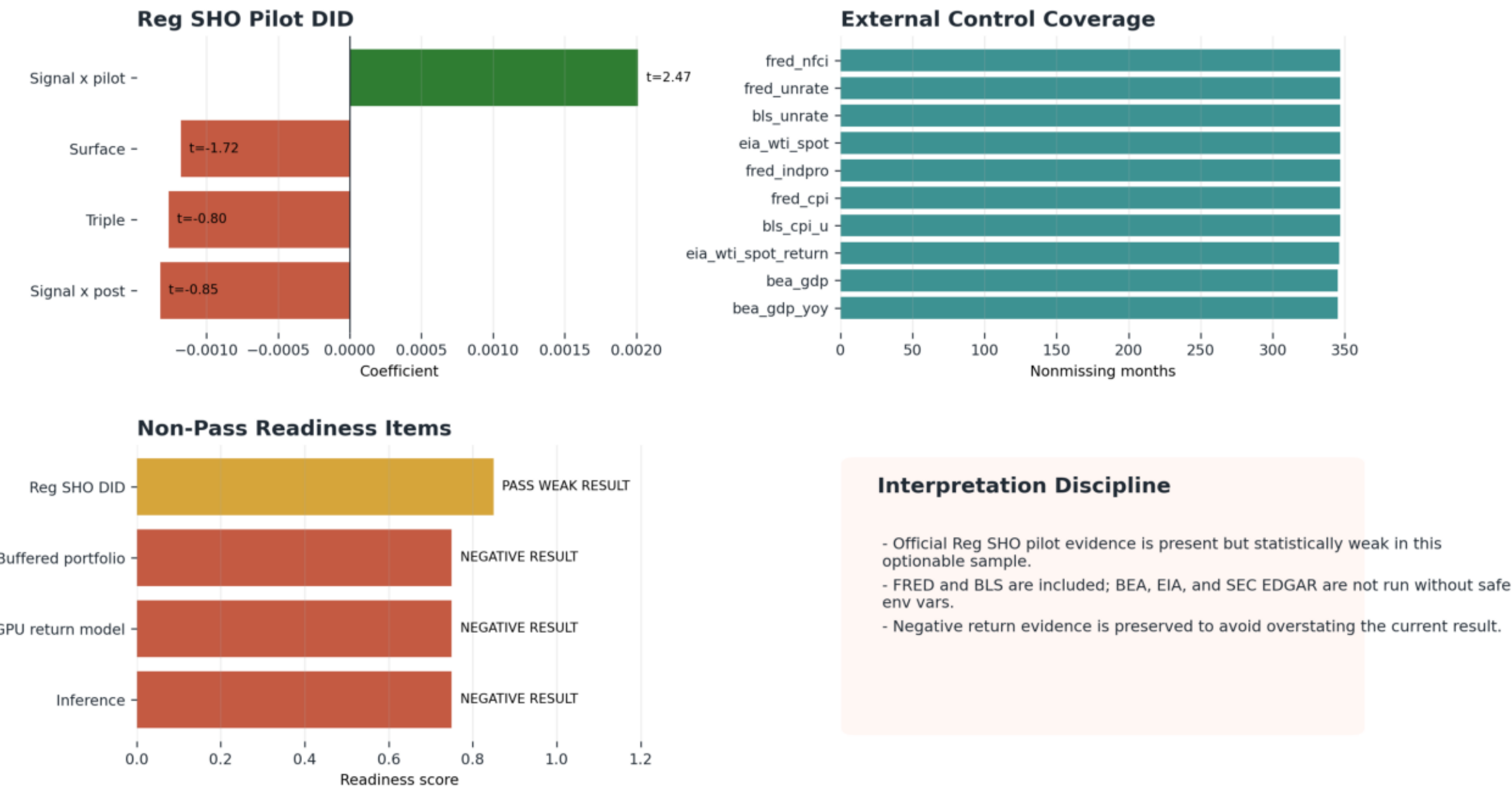


Figure 5. Mechanisms and Readiness

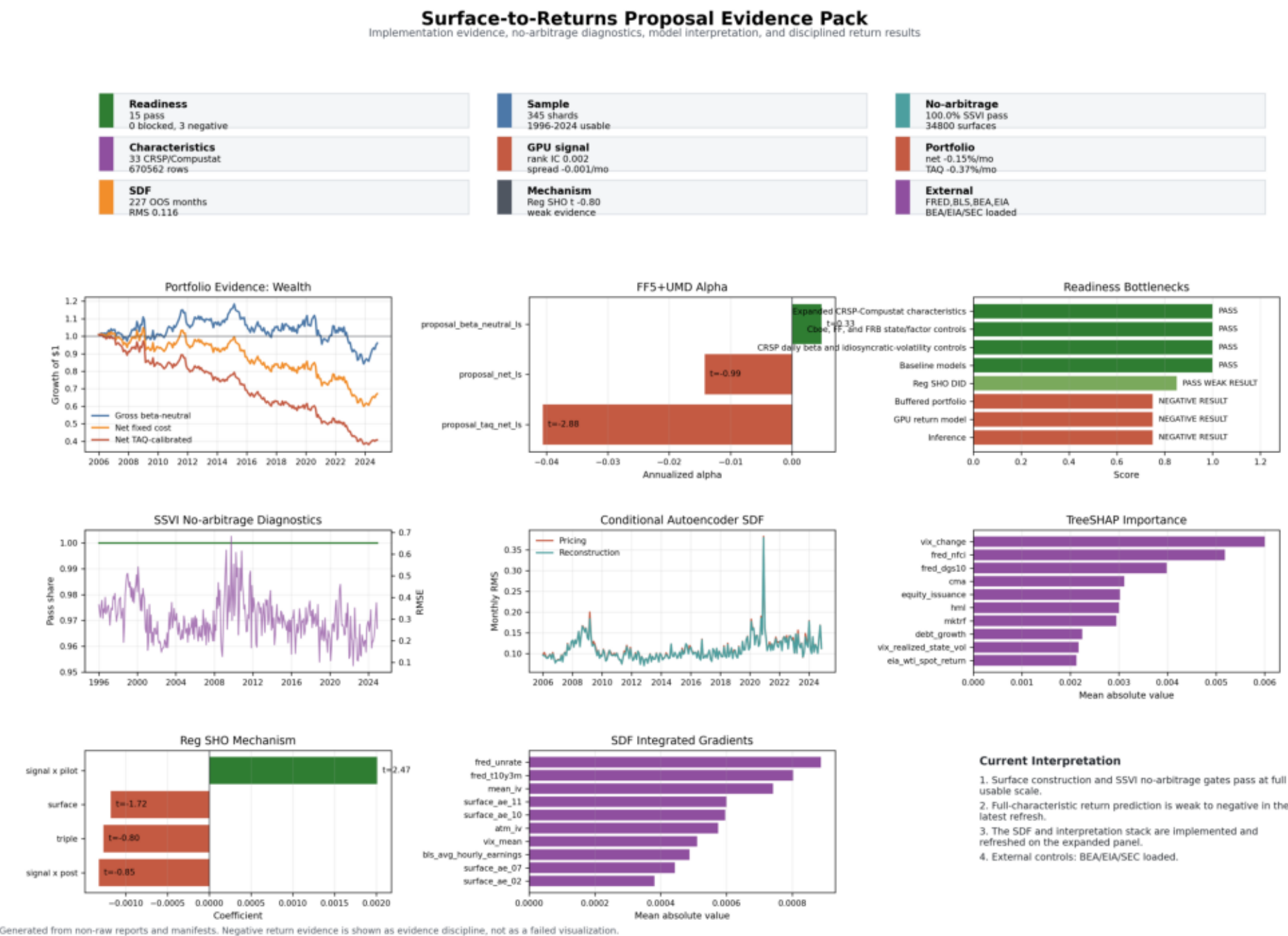
Reg SHO mechanism evidence, external controls, non-pass readiness items, and interpretation discipline.

Figure 5. Mechanism, External Controls, and Remaining Readiness Items



Proposal Evidence Pack

Detailed one-page view of implementation status, diagnostics, model evidence, and boundaries.



Generated from non-raw reports and manifests. Negative return evidence is shown as evidence discipline, not as a failed visualization.

Evidence Discipline

The package separates implemented infrastructure from claims the current evidence does not support.

Implemented

- Scaled WRDS-backed extraction and linkage through the usable 1996-2024 sample.
- SVI and SSVI no-arbitrage diagnostics pass at full usable scale.
- Characteristics, state controls, TAQ costs, IBES, Reg SHO, SDF, SHAP, and inference artifacts exist.
- Figures, report, deck, and PDF are generated from non-raw outputs and manifests.

Not Overclaimed

- GPU walk-forward return model: NEGATIVE RESULT
- Sector-balanced beta-hedged buffered portfolio: NEGATIVE RESULT
- Factor alphas, HJ/GRS, Newey-West, bootstrap inference: NEGATIVE RESULT
- Reg SHO pilot mechanism test: PASS WEAK RESULT